Vidyavardhini's College of Engineering & Technology

Department of Computer Science Engineering (Data Science)

Academic Year: 2026-27

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2026-27

Course: Wireless and Mobile Communication
Course code: **Year: TE**
SEM: V

Experiment No. 4
Aim: To perform Wi-Fi network scanning using a mobile application and analyze the SSID, RSSI, Wi-Fi channel, and frequency band of nearby wireless networks.
Name: Meet Mehta
Roll Number: 30
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by

Name of Faculty : Mrs. Sonali Sable
Signature : **Date :**

Aim:

To perform Wi-Fi network scanning using a mobile application and analyze the SSID, RSSI, Wi-Fi channel, and frequency band of nearby wireless networks.

Theory:

1. Wi-Fi Network Scanning

Wi-Fi network scanning is the process of detecting wireless networks available in the surrounding area using a Wi-Fi-enabled device.

A Wi-Fi Analyzer application can scan nearby networks and display information such as SSID, RSSI, channel number, and frequency band.

2. SSID

SSID (Service Set Identifier) is the name used to identify a Wi-Fi network.

For example, when a smartphone displays Wi-Fi networks such as Home_WiFi, College_WiFi, or JioFiber, these names are SSIDs.

3. RSSI

RSSI (Received Signal Strength Indicator) represents the received strength of a Wi-Fi signal.

It is generally expressed in dBm.

- A value closer to 0 dBm indicates a stronger signal.
- A more negative value indicates a weaker signal.

For example, −40 dBm generally represents a stronger signal than −80 dBm.

4. Wi-Fi Channel

A Wi-Fi channel is a portion of the radio-frequency spectrum used by a wireless network for communication.

For example, in the 2.4 GHz band, Wi-Fi networks commonly operate on channels such as 1, 6, and 11.

When several nearby networks operate on the same or overlapping channels, interference may occur.

5. Frequency Band

Wi-Fi networks commonly operate in frequency bands such as:

- 2.4 GHz – provides better coverage but is more prone to interference.
- 5 GHz – generally provides higher data rates and more available channels, but has shorter coverage compared with 2.4 GHz.

The Wi-Fi Analyzer application can be used to identify the frequency band used by nearby networks.

6. Channel Occupancy

Channel occupancy indicates how many nearby Wi-Fi networks are operating on a particular channel.
A channel with many nearby networks is considered more crowded, while a channel with fewer networks is less crowded.

Procedure

Step 1: Install the Wi-Fi Analyzer Application
Install a suitable Wi-Fi Analyzer application on the smartphone.

Step 2: Open the Application
Open the installed Wi-Fi Analyzer application.
Observe the main screen and the available Wi-Fi scanning or analysis options.

Step 3: Enable Wi-Fi
Enable Wi-Fi on the smartphone.
Make sure the smartphone is able to detect nearby wireless networks.

Step 4: Provide Required Permissions
Provide the permissions requested by the application, such as Location permission, if required by the smartphone or application.

Step 5: Start Wi-Fi Scanning
Start the Wi-Fi scan in the application to detect nearby wireless networks.

Step 6: Observe Detected Networks
Observe the list of Wi-Fi networks detected by the application.
For each detected network, observe the available information such as:

1. SSID
2. **RSSI**
3. **Channel number**
4. **Frequency band**

Step 7: Record the SSID

Select or note the SSID (name) of each detected Wi-Fi network.

Record the SSID in the observation table.

Step 8: Record RSSI

Observe the RSSI/signal strength of each detected network.

Record the RSSI value in dBm.

Step 9: Record Channel Number

Observe the Wi-Fi channel number used by each detected network.

Record the corresponding channel number in the observation table.

Step 10: Record Frequency Band

Observe the frequency band used by each Wi-Fi network, such as 2.4 GHz or 5 GHz.

Record the frequency band in the observation table.

Step 11: Open Channel Analysis

Open the channel graph or channel analysis view available in the Wi-Fi Analyzer application.

Observe how the detected networks are distributed across different Wi-Fi channels.

Step 12: Observe Channel Occupancy

Observe the number of nearby Wi-Fi networks operating on each channel.

Note which channels have a large number of networks and which have fewer networks.

Step 13: Identify Crowded and Less-Crowded Channels

Identify the most crowded and least crowded Wi-Fi channels based on the displayed channel analysis.

Step 14: Compare the Detected Networks

Compare the RSSI values, channels, frequency bands, and channel occupancy of the detected Wi-Fi networks.

Observe which networks have stronger or weaker signals and which channels are more crowded.

Step 15: Record Observations and Conclusion

Record all observations in the observation table.

Based on the obtained results, write the conclusion regarding the detected Wi-Fi networks, their signal strengths, frequency bands, and channel usage.

OUTPUT:


Access Points
2.4
5
6
GHz
PRO
IN RANGE 11
by Strength
OPPO Reno13 5G t99u
WiFi 4
c6:96:0d:33:25:64 4m
2437MHz (20MHz) CH 6
WPA3
-53
dBm
Vivo
WiFi 4
aa:51:4e:f8:54:b7 6m
2437MHz (20MHz) CH 6
WPA2
-55
dBm
VCET
WiFi 4
e8:48:b8:3d:3b:02 16m
2412MHz (20MHz) CH 1
WPA2 [TP-Link]
-64
dBm
sonali
WiFi 4
72:3a:51:f1:53:41 34m
2472MHz (20MHz) CH 13
WPA2
-71
dBm
GSVCET 2.4G
WiFi 4
64:29:43:b7:0b:c0 35m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-71
dBm
EXAMSECTION 2.4G
WiFi 4
64:29:43:b7:0b:f0 35m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-71
dBm
office_conference
WiFi 4
10:27:f5:37:55:b2 49m
2462MHz (20MHz) CH 11
WPA2 [TP-Link]
dBm
Gold & Silver Tokens at
Lowest Fees
Trade Now


Access Points
2.4
5
6
GHz
PRO
IN RANGE 1
by Strength
GSVCET 5G
WiFi 5
64:29:43:b7:0b:c8
37m
5745MHz (80MHz)
CH 149(155)
WPA2
26Mbps
[D-Link]
-79
dBm

Observation Table:

Sr. No.	SSID	RSSI (dBm)	Channel No.	Frequency Band
1	OPPO Reno13 5G t99u	-53	6	2.4 GHz
2	Vivo	-55	6	2.4 GHz
3	VCET	-64	1	2.4 GHz
4	sonali	-71	13	2.4 GHz
5	GSV CET 2.4G	-71	1	2.4 GHz
6	EXAMECTION 2.4G	-71	1	2.4 GHz
7	GSVCET 5G	-79	149 (155)	5 GHz
8	iBall-Baton	-90	6 (4)	2.4 GHz

Note: Record the observations for each detected Wi-Fi network individually in the observation table, ensuring that all entries match the output displayed by the Wi-Fi Analyzer application.

Conclusion: Thus, Wi-Fi network scanning was successfully performed using the Wi-Fi Analyzer application. Several nearby networks were detected in the 2.4 GHz and 5 GHz bands. The RSSI values varied for different networks, indicating different signal strengths. Multiple networks were observed on channel 1, making it a relatively crowded channel in the scanned area. The scan successfully provided information about SSID, RSSI, channel number and frequency band.

VIVA Questions

What is Wi-Fi network scanning?

Wi-Fi network scanning is the process of detecting wireless networks available in the surrounding area.

What is SSID?

SSID stands for Service Set Identifier. It is the name used to identify a Wi-Fi network.

What is RSSI?

RSSI stands for Received Signal Strength Indicator. It represents the received strength of a Wi-Fi signal.

In which unit is RSSI measured?

RSSI is generally measured in dBm.

What is channel occupancy?

It indicates how many nearby Wi-Fi networks are operating on a particular channel.